

Assignment 1

Statistical Inference 2026-27

September 24, 2026

Discussion among yourselves is permitted and encouraged. AI usage is permitted but discouraged.

Problem 1: Casella-Berger Theorem 6.2.28 (5 points)

Prove the following result: *If a minimal sufficient statistic exists, then any complete statistic is also a minimal sufficient statistic.*

Problem 2: MLE, the Cramér–Rao Bound, and UMVUE (10 points)

Let $X = (X_1, \dots, X_n)$ have likelihood $L(\theta; x)$, where θ is a scalar parameter, and define

$$U(\theta; x) = \frac{\partial}{\partial \theta} \log L(\theta; x).$$

A student makes the following argument:

The maximum likelihood estimator $\hat{\theta}_{MLE}$ satisfies $U(\hat{\theta}_{MLE}; X) = 0$. The equality condition for the Cramér–Rao lower bound (CRLB) is

$$U(\theta; X) = a(\theta)(W(X) - \theta),$$

where $a(\theta) \neq 0$. Substituting $\theta = \hat{\theta}_{MLE}$ gives $W = \hat{\theta}_{MLE}$. Therefore every MLE attains the CRLB and is a uniformly minimum variance unbiased estimator (UMVUE).

1. Identify the logical flaw in this argument.
2. Give a counterexample in which the MLE of θ is not the UMVUE. Derive the MLE and justify why it is not the UMVUE.
3. State sufficient conditions under which an MLE is unbiased, attains the CRLB, and is therefore the UMVUE.

Problem 3: EM with Imperfect Blood Typing (10 points)

Let p_A, p_B, p_O be allele frequencies, with $p_A + p_B + p_O = 1$. Assume the genotype probabilities are

Genotype	AA	AO	BB	BO	AB	OO
Probability	p_A^2	$2p_A p_O$	p_B^2	$2p_B p_O$	$2p_A p_B$	p_O^2

The corresponding true blood types are A, A, B, B, AB, O .

A laboratory reports the correct blood type with probability $1 - \varepsilon$. Otherwise, it reports each of the other three types with probability $\varepsilon/3$. Assume $\varepsilon = 0.10$ is known and laboratory errors are independent across individuals, conditional on their true blood types. You observe only the reported blood types.

1. **Derive and implement EM.** Treat the six genotype counts as latent. Derive the conditional genotype probabilities given a reported blood type and obtain the allele-frequency updates. Implement the algorithm in R or Python.
2. **Check your implementation.** Simulate $n = 1000$ independent observations using

$$(p_A, p_B, p_O) = (0.30, 0.15, 0.55).$$

Run EM from three distinct starting points in the interior of the probability simplex. Plot the observed-data log-likelihood over iterations and compare the final estimates with direct numerical maximization of that likelihood.

3. **Investigate the cost of ignoring error.** Across 300 simulated datasets of size $n = 1000$ from the model in part 2, compare your estimator with an estimator that incorrectly assumes $\varepsilon = 0$. Report the empirical bias and root mean squared error for each allele frequency. Explain the direction of the differences using the model's reported-type probabilities.
4. **Stress-test the method.** Repeat the comparison between EM and direct maximization in part 2 using

$$(p_A, p_B, p_O) = (0.02, 0.18, 0.80).$$

Discuss convergence, estimates near the boundary, and whether a small change in log-likelihood necessarily implies accurate parameter estimates.

Submit a short write-up containing the requested plots and numerical summaries, and a short explanation of your findings. DO NOT print and submit your code. Instead upload it on onedrive in [this link](#) as A1_Firstname.R.